Gravitational Compression and Microscopic Core Closure: A Mechanical Squeezing Formulation of the Slowik Hypothesis

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Abstract

We reformulate the gravitational collapse framework by replacing thermal assumptions with direct mechanical compression and hydrodynamic squeezing[span$_3$]($start_span$)[$span_3$](end_span).$Buildingupon$ $M_\odot$, high-density regimes of QCD and electroweak scales ($\rho_{\text{core}} \approx 5.27 \times 10^{30}$ kg m^{-3}) confine the entire mass budget within a sub-meter core ($r_{\text{core}} \approx 0.63$ m), completely bypassing prior energy-budget obstructions and enabling structural core closure[span$_6$]($start_span$)[$span_6$](end_span).

1 Introduction

Previous relativistic collapse analyses and foundational formulations (such as Stage II and Revision III developments) encountered an energy-budget obstruction when treating high-energy Standard-Model thresholds as macroscopic, thermally equilibrated spheres[span$_7$]($start_span$)[$span_7$](end_span)[s

In this work, we shift the paradigm: we replace the thermal assumption with direct **mechanical compression** (hydrodynamic squeezing)[span$_9$]($start_span$)[$span_9$](end_span).$Gravityactsasanon-linearcompressionoperatorconcentratingenergyintoadeepcoreratherthanaglobalthermalbubble$[$span_1$0]($start$

2 The Hydrodynamic Squeezing Model

Consider a reference mass based on the Package Formula[span$_1$1]($start_span$)[$span_1$1](end_span)[$span_1$2]($start_sp$ $2.8M_\odot \approx 5.57 \times 10^{30}$ kg(1) The corresponding Schwarzschild radius is:

$$R_S = \frac{2GM}{c^2} \approx 8.27 \text{ km} \tag{2}$$

Defining the core density corresponding to extreme mechanical compression as $\rho_{\text{core}} \approx 5.27 \times 10^{30}$ kg m^{-3}, the required volume is[span$_1$3]($start_span$)[$span_1$3](end_span) : $V_{\text{core}} = \frac{M}{\rho_{\text{core}}} \approx 1.057$ m^3(3)

The resulting effective core radius evaluates to[span$_1$4]($start_span$)[$span_1$4](end_span) : $r_{\text{core}} = \left(\frac{3V_{\text{core}}}{4\pi}\right)^{1/3} \approx 0.632$ m(4)

3 Physical Implications

1. **Mass-Budget Compliance:** The entire $2.8M_\odot$ mass fits inside a ~ 63 cm sphere, fully satisfying the mass budget without requiring external inputs[span$_1$5]($start_span$)[$span_1$5](end_span).

1. **Sub-Horizon Localization:** The core scale resides deep inside the event horizon (8.27 km), operating safely within strong-field dynamics[span$_1$6]($start_span$)[$span_1$6](end_span).

1. **Universal Scaling:** This transition opens a pathway for discrete geometric closure operations, aligning with the broader relational framework[span$_1$7]($start_span$)[$span_1$7](end_span).

References

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